

3D-SGRL introduces 3D benchmarks and a subequivariant transformer that outperforms SOTA

Runfa Chen*, Jiaqi Han*, Fuchun Sun, Wenbing Huang

1 Dept. of Comp. Sci. & Tech., Institute for AI, BNRist Center, Tsinghua University · 2 THU-Bosch JCML Center · 3 Gaoling School of Artificial Intelligence, Renmin University of China · 4 Beijing Key Lab of Big Data Management and Analysis Methods

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- New 3D morphology-agnostic RL benchmarks with full DoFs
- SubEquivariant Transformer (SET) enforcing $O_g(3)$ symmetry
- Superior single-task, multi-task, and zero-shot performance

Code & videos: <https://alpc91.github.io/SGRL/> Keywords: RL · Equivariance · Transformers · GNNs · 3D Locomotion

* Equal contribution · Contact: fcsun@mail.tsinghua.edu.cn, hwenbing@126.com

Subequivariance in 3D improves exploration and generalization

- Prior morphology-agnostic RL mostly constrained to 2D, fixed start/goal, limited exploration.
- We propose 3D-SGRL: agents operate in full 3D with arbitrary starts and targets.
- Inject geometric symmetry (subequivariance) into policy and Q to generalize across directions.
- Introduce SubEquivariant Transformer (SET) enabling expressive, symmetry-preserving message passing.
- Consistent gains over SOTA baselines across tasks and generalization settings.

Space

Full 3D

Policy

$O_g(3)$ -subequivariant

Model

Transformer-based

3D benchmarks expand DoFs and targets beyond 2D constraints

(a) 2D Planar Locomotion — Image Placeholder

(b) 3D Subequivariant Locomotion — Image Placeholder

- 3D agents gain more DoFs and can move/turn arbitrarily.
- Policies must generalize across orientations and targets.

Encoding $O_g(3)$ symmetry with attention improves cross-task performance

- Benchmarks: Practical 3D locomotion with arbitrary starts/targets; increased joint DoFs.
- Method: SET enforcing rotation/translation symmetry under gravity via $O_g(3)$ subequivariance.
- Results: Significant outperformance in single-/multi-task and zero-shot; extensive ablations.

Graph-structured shared policies enable morphology-agnostic transfer

- Agents modeled as graphs $G=(V,E)$ where nodes=limbs, edges=joints.
- Shared policy π_{θ} over multiple morphologies using GNNs/Transformers.
- Actor-critic (e.g., TD3) with limb-level states and torques in continuous control.

Takeaway: graph-structured, shared policies enable morphology-agnostic transfer.

Subequivariance captures gravity while preserving 3D geometric consistency

- $E(3)$ -equivariance: $f(g \cdot Z, h) = g \cdot f(Z, h)$ for 3D transformations.
- Subequivariance $O_g(3)$: symmetry restricted to gravity-aligned rotations/reflections.
- Allows capturing gravity effects while preserving necessary geometric generalization.

Mathematical definitions & constructions — Placeholder

Larger state/action spaces plus $O_g(3)$ symmetry improve generalization

State Space

- 2D-Planar: x-oz projection; fixed +x heading; single direction.
- 3D-SGRL: full 3D; random init; arbitrary target directions.

Action Space

- 2D-Planar: 1 actuator/DoF per joint.
- 3D-SGRL: 3 actuators/DoFs per joint.

Symmetry

- 2D-Planar: No explicit symmetry handling.
- 3D-SGRL: Gravity g and target d ; $O_g(3)$ subequivariance.

Table 1 — Comparison Placeholder

SET preserves $O_g(3)$ structure and yields invariant Q and actions

- Input processing: split limb states into directional vectors Z_i and scalars h_i ; include g and target d .
- SubEquivariant Transformer (SET): symmetry-preserving self-attention over morphological graph.
- Actor & Critic heads: $O_g(3)$ -consistent action and invariant Q-function.

Input Processing

Inputs: raw states, g , target $d \rightarrow$ Outputs: $\{Z_i \in \mathbb{R}^{3 \times 6}, h_i \in \mathbb{R}^{13}\}$; translation- and height-aware features.

- What: separate steerable vectors vs invariants; add g, d .
- Why: ensures rotations act only on Z_i ; enables $O_g(3)$ handling.

split vectors/scalars

normalize (trans/height)

SET Layers ($\times L$)

Inputs: $\{Z_i, h_i, g, d\} \rightarrow$ Outputs: equivariant vectors + invariant matrices via invariant Q/K and attention.

- What: build invariant queries/keys; mix channels into M_i .
- Why: attention weights are symmetry-invariant; updates stay equivariant.

Actor & Critic Heads

Inputs: final equivariant vectors/invariants $\rightarrow$ Outputs: action a , invariant Q-function

Structured O_g(3)-aware inputs enable symmetry-consistent learning

- Z_i ($R^{3 \times 6}$): position p_i , velocity v_i , angular vel ω_i , joint axes x_i, y_i, z_i .
- Note: Z_i carries all steerable geometry; rotates with $O_g(3)$ to preserve consistency.
- h_i (R^{13}): joint angles (κ, ζ, δ), ranges, limb type (one-hot).
- Note: h_i is invariant; isolates scalars from rotations.
- Target direction d from torso to assigned target (xy-plane, lifted to R^3).
- Note: d provides task guidance; treated as steerable to align with symmetry.

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Invariant attention over mixed vectors yields equivariant updates

- Preprocess to translation invariants and height features.
- Compute $O_g(3)$ -invariant value matrices M_i from mixed vector channels and $\{g, d\}$.
- Self-attention via invariant queries/keys from M_i ; aggregate equivariant and invariant messages.
- Stack L layers for expressive, symmetry-compliant message passing.

SET attention equations & pipeline — Placeholder

- Step 1: compute invariant M_i to mix vector/scalar channels.
- Step 2: form invariant Q/K from $\text{vec}(M_i)$ to obtain symmetry-stable weights.
- Step 3: apply α_{ij} to equivariant values to update Z and invariant values to update M .
- Step 4: layer-normalize and stack L times for expressive fusion.

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Equivariant torque readouts and invariant critics stabilize control

- Actor: readout equivariant torque vector $T_i \in R^{3 \times 1}$; project onto joint axes $\rightarrow a_i$ (invariant scalars).
- Critic: invariant Q via linear map over final invariant matrices M_i (actions concatenated to inputs).
- Separate SETs for actor and critic.

Actor readout & projection · Critic invariance — Placeholder

Caption: Equivariant torque readout preserves orientation; axis projection yields invariant actions, while invariant readout maps to Q .

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3D variants and curated collections enable rigorous cross-domain evaluation

- 3D counterparts of Hopper, Walker, Humanoid, Cheetah with multiple variants.
- Collections: In-domain (Hopper++, Walker++, Humanoid++, Cheetah++); Cross-domain (WHH++, CWHH++).
- 20% variants held out for zero-shot testing.
- Extended cheetah: full-cheetah (14 limbs) for stable 3D locomotion.

Variant lists & structures — Placeholder

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Randomized goals and full DoFs create challenging, realistic tasks

State & Init

- Random torso orientation; random destination within radius R.
- v2-variants: $R \in [10\text{ m}, 20\text{ m}]$; default $R=10\text{ km}$ for fewer turns.

Action

- Torso joint type $\rightarrow$ free; 3 actuators/axes per joint (full DoFs).

Termination & Reward

- Termination: as prior benchmarks; cheetah ends on fall/squat.
- Rewards: alive bonus; locomotion toward target; control cost; forward reward (except Hopper).

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SET consistently leads multi-task training across all collections

- Baselines: SMP [Huang et al., 2020], SWAT [Hong et al., 2021]; Monolithic (single-task ref).
- Algorithm: TD3 for all methods; identical inputs; 3 seeds; ep len 1000.
- Metrics: multi-task (seen), zero-shot (unseen variants), v2 transfer, single-task.

Metric definitions

- Multi-task (seen): train on 80% variants; evaluate mean episode return on these seen variants over seeds.
- Zero-shot (unseen): evaluate trained policies on held-out 20% morphologies; report mean $\pm$ SE over 3 seeds, each over 100 rollouts.
- v2 transfer: pretrain at $R=10\text{ km}$, then fine-tune/evaluate on $R \in [10\text{ m}, 20\text{ m}]$; report episode returns.
- Single-task: train and evaluate on one morphology; returns are summed rewards per episode; report mean $\pm$ SE over seeds.

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SET dominates multi-task training across all collections

- SET outperforms SWAT and SMP across all environment collections.
- Baselines struggle due to enlarged exploration space in 3D.

Figure 3 — Training curves (multi-task) Placeholder

Consistent and significant gains with subequivariance.

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Transfer training boosts SET on v2 variants

- Two-stage: pretrain ($R=10\text{ km}$) $\rightarrow$ transfer ($R=10\text{--}20\text{ m}$).
- SET improves further; SWAT shows limited gains on Humanoid++.

Figure 4 — v2-variants curves Placeholder

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SET outperforms monolithic and SMP/SWAT on single-task cheetah and humanoid

- SET surpasses Monolithic, SMP, SWAT on cheetah/humanoid cases.
- Transformer expressiveness + subequivariance boosts stability and returns.

Figure 5 — Single-task curves Placeholder

SET achieves higher zero-shot returns in both in-domain and cross-domain

- Held-out morphologies: SET achieves higher returns in both in-/cross-domain.

Table 2 — Zero-shot results Placeholder

SET matches or exceeds HN methods without normalization assumptions

- Heading Normalization (HN) offers heading-equivariance but assumes consistent forward direction.
- SET comparable or better; more generalizable without agent-specific normalization.

Figure 6 — Multi-task vs HN Placeholder

HN is fragile to heading bias; SET generalizes better on zero-shot

HN Bias Test (Single-Task)

- Adding 180° bias degrades Monolithic+HN and SWAT+HN sharply.
- SET remains robust without normalization assumptions.

Table 3 — Bias test Placeholder

Zero-Shot vs HN

- SET shows stronger generalization on CWHH++ test variants.

Table 4 — Zero-shot vs SWAT+HN Placeholder

Removing gravity/direction inputs significantly reduces returns across agents

- SET\g, SET\z: gravity/height matter more for humanoid than cheetah.
- SET\gd: removing directional task info hurts.
- SET in invar: dropping vectors to scalars degrades notably.
- SET out invar: worse than full SET; equivariant output beneficial.

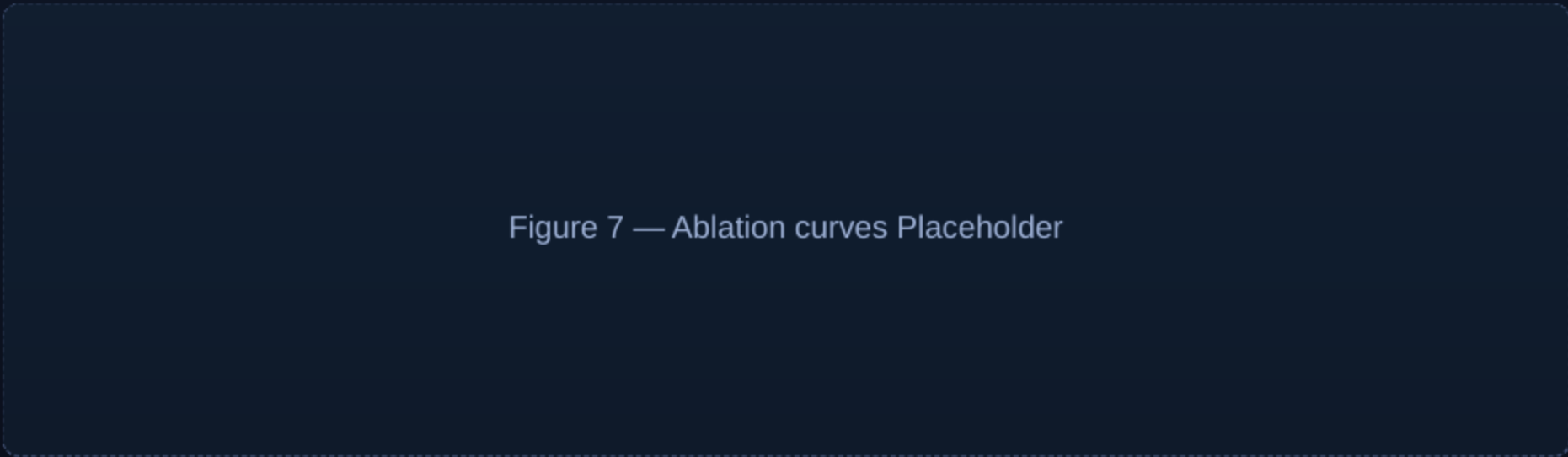


Figure 7 — Ablation curves Placeholder

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Equivariance with attention scales shared policies across diverse morphologies

- Equivariance + attention enable shared policies across diverse 3D morphologies.
- Simple MLP+HN can be strong in specific single-task cases; our focus is multi-task generality.
- 3D-SGRL benchmarks advance realistic evaluation for morphology-agnostic RL.

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Key open problems for scaling subequivariant RL in 3D

- How to extend subequivariance beyond gravity (e.g., multiple external fields)?
- Scalability of SET to even larger multi-agent, multi-morphology regimes.
- Bridging sim-to-real with 3D-SGRL policies under real-world sensing/noise.
- Automated discovery of symmetry structure from data rather than design.
- Task-conditioned generalization to diverse goals beyond directional guidance.

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References and resources for 3D-SGRL and SET

- Huang et al., 2020 (SMP): “One Policy to Control Them All: Shared Modular Policies for Agent-Agnostic Control.”
- Hong et al., 2021 (SWAT): “Structure-Aware Transformer Policy for Inhomogeneous Multi-Agent Reinforcement Learning.”
- Mnih et al., 2015/2016; Silver et al., 2016; Schulman et al., 2017; Fujimoto et al., 2018; Battaglia et al., 2018; Vaswani et al., 2017.
- Cohen & Welling, 2016/2017; Thomas et al., 2018; Fuchs et al., 2020; Satorras et al., 2021; Han et al., 2022.
- Project: <https://alpc91.github.io/SGRL/> · Paper: ICML 2023, PMLR 202.

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